

The Impact of the Internet of Things



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How would the various new-age devices connected to the Internet of Things 'talk' to each other?

The communication link of the IoT depends on the technology that is available on smartphones which is, for the most part, Wi-Fi and for the sensor devices, it is Bluetooth. It is not just about connecting everything and having them communicate because that does not answer the question of what is it saying and who does it talk to.

Technologies like LTE are going to be how devices talking to each other would happen. On the industry's side, it will definitely be a combination of existing technologies like WPAN as well as lower spectrum technologies such as 2G or 2.5G, because the cost and amount of data is very different with respect to the consumer.

That remains a challenge since there are so many protocols, different types of devices that do not talk to each other very well. We should look past this challenge and look at what makes sense. For example, it does not make sense for a projector to talk to a chair. But it makes sense for a projector to talk to a smart phone. Some of these challenges are going to be addressed.

If devices are going to be connected, we need to have a language. Maybe it is at a level above the physical layer, but some of these applications can define certain behaviours or how they can talk to each other. That is exactly where the IEEE Standards Association or some other standards body can come in, and say this is very ripe for standardisation.

The advent of nanotechnology will enable smarter sensors and actuators which can also capture context information along with data. Having tiny wireless sensors which can operate on ultra-low power to be able to work for 10-20 years and communicate wirelessly is one key element to achieve this and Bosch is also active on this.

The IoT relies on inexpensive, small sensors. These sensors need to be power efficient, and able to run on batteries that last for 10 years. They also need to be design efficient for integration into minute things that could cost less than \$1. But not every thing will contain a tiny microcontroller running off a battery.